ROLL NO: | | | | | | | | | | |

(To fill the roll no. is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION-2024 - 2025

DIPLOMA (ME/CE/EE/CSE)

Semester: II

SUBJECT CODE: DAS201

Mathematics-II

Time: 3 Hours **Maximum Marks: 60**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION- A

NOTE: Attempt all parts of the following questions. **(12 Marks)**

Q. 1. Answer the following questions: **(1 × 6 = 6 Marks)**

(i) Define symmetric relation.
(ii) Define the set.
(iii) Give an example of reflexive relation which is not transitive.
(iv) If A = {2, 4 ,6} and B = {1, 5}. Find A× B.
(v) Every function is a relation **(T/F).**
(vi) Define one-one function.

Q. 2. Answer the following questions: **(1 × 6 = 6 Marks)**

(i) What is continuous function?
(ii) Every continuous function is differentiable?
(iii) Evaluate $\int_1^2 x\,dx$.
(iv) What is the domain of the function?
(v) Give an example of null matrix.
(vi) Differentiate the following function with respect to x e^{1993x}dx.

SECTION-B

NOTE: Attempt all the questions. **(6 × 2 × 2 = 24 Marks)**

Q.3.(i)(a) Solve the system of equation:

x+2y = 4
2x-y = 3.

OR

(b) Show that the function f: R → R defined by f(x) = $3x^3 + 5$ for all x ∈ R is a bijection.

(ii) (a) Show that f(x) = |x| is not differentiable at x=0.

OR

(b) Let P = $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, Q = $\begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix}$. $If\ P + Q = \begin{bmatrix} 50 & 70 \\ -90 & 110 \end{bmatrix}$. Determine the matrix P.

Q.4. (i) (a) If $\int_1^a (3x^2 + 2x + 1)dx$ = 11, find real values of a.

OR

(b) Given M = $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, find M-2I, where I is an identity matrix of order 2.

(ii) (a) Show that the function f(x) = 2x - |x| is continuous at x=0.

OR

(b) Find the derivative of $x^5 e^{5x}$.

SECTION-C

NOTE: Attempt all questions. **(12× 2 = 24 Marks)**

Q.5. (a) Find the inverse of the matrix A = $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

OR

(b) Evaluate $\int_0^{\pi/2} \frac{cos\ x}{1+cos\ x+\sin x}$dx.

Q.6. (a) Find the area of the region bounded by the curve y^2 = 2y-x and the x-axis.

OR

(b) Prove that the relation R on the set Z of all integers numbers defined by (x, y) ∈ R if and only if x-y is divisible by n, is an equivalence relation on Z.